

# **Day 4**

# **Multilingual learning**

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# Pretrained multilingual models

A typical approach in NLP: “we trained a multilingual model on N (typologically diverse) languages”

This description hides a lot of methodological details, e.g.:

- How much training data from each?
- How was the sampling done?
- In what order?

# Pretrained multilingual models

Some of the modeling decisions:

- **Tokenizer & vocabulary:** a shared subword vocabulary vs. separate
- **Embeddings & weights:** fully shared (the default) vs. adding language-specific parameters (e.g., 'adapters').
- **Cross-lingual signal:** implicit (shared vocab + joint monolingual training, nothing pairing the languages) vs. explicit (translation pairs).

# Pretrained multilingual models

mBERT (Devlin et al., 2019):

- 104 languages
- Shared WordPiece tokenizer with 110k vocabulary
- Low-resource languages intentionally oversampled

# Transfer learning

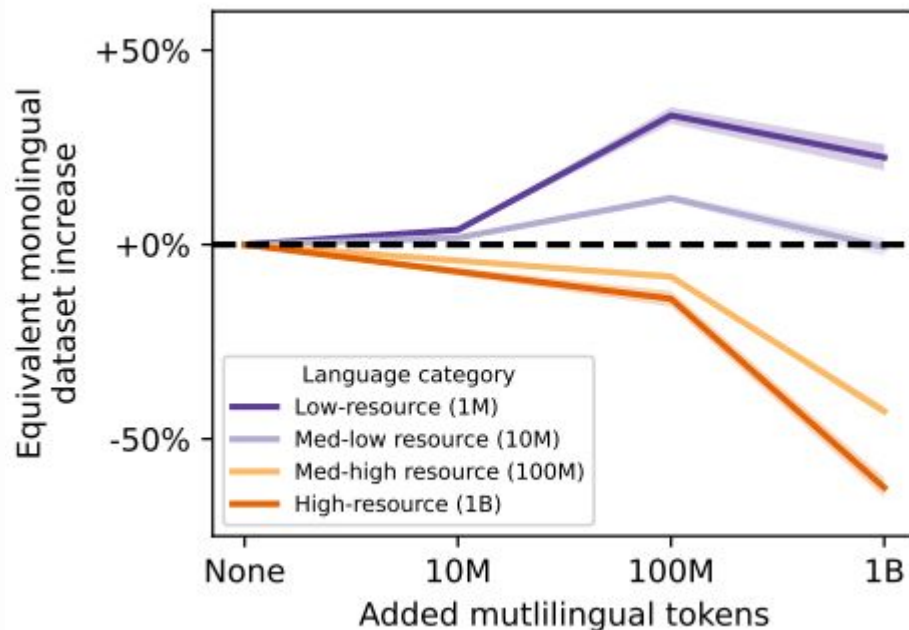
- Classic transfer learning paradigm: In multilingual models, performance on low-resource languages can increase thanks to (positive) cross-lingual transfer from high-resource ones.
- But performance on high-resource languages might drop, once low-resource languages are added, due to (negative) cross-lingual transfer.

# Curse of multilinguality

- Conneau et al. (2020) call this trade-off *curse of multilinguality*
  - At *fixed model capacity*, per-language capacity decreases as we increase the number of languages
- Chang et al. (2024) reproduce the 'curse' on a large number of languages (252) and models (10,000+)

# Curse of multilinguality

*“massively multilingual pretraining may not be optimal for any languages involved, but <...> more targeted models can significantly improve performance”*



Chang et al. (2024)

# Multilingual BERT

- Transfer occurs even across scripts
- Works best between typologically similar languages
- Model can find translation pairs

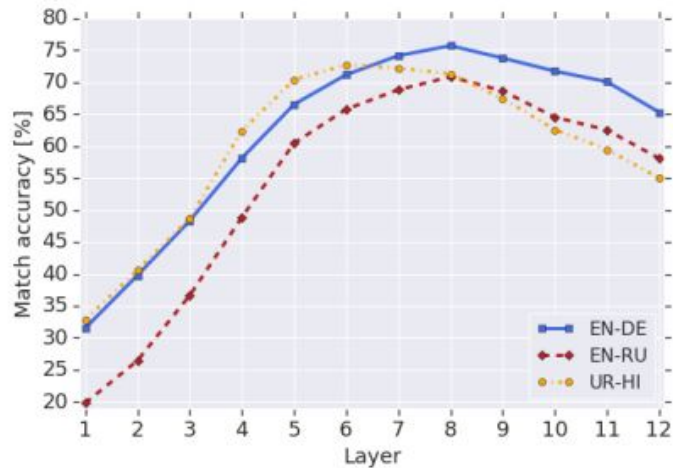


Figure 3: Accuracy of nearest neighbor translation for EN-DE, EN-RU, and HI-UR.

Pires et al. (2019)



# Multilingual 'BERTology'

## Beto, Bentz, Becas: The Surprising Cross-Lingual Effectiveness of BERT

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## CROSS-LINGUAL ABILITY OF MULTILINGUAL BERT: AN EMPIRICAL STUDY

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## Emerging Cross-lingual Structure in Pretrained Language Models

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# Shared or separate representations?

- **General finding:** both shared and language-specific components
- Libovický et al (2020) : representations “strongly carry information about language identity”
- One desideratum at that time: a truly cross-lingual space where you can easily switch from one language to another without even training on that language

# Causes of 'multilinguality'

A combination of architectural choices and linguistic properties:

- Shared positional/segment embeddings
- Shared special tokens
- Cross-lingual signal ("code-switching" in training)
- Similar word order

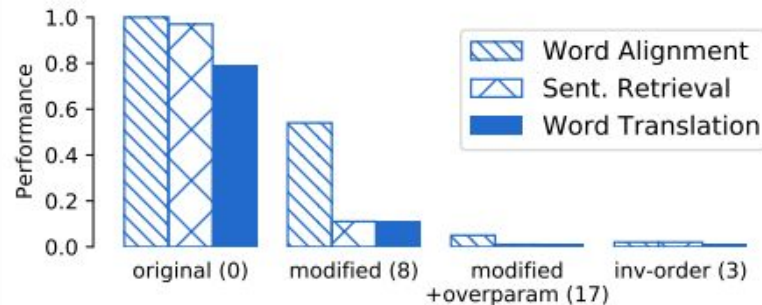


Figure 1: Multilinguality in our BERT model (0) is harmed by three architectural modifications: lang-pos, shift-special, no-random (8); see §2.3 for definitions. Together with overparameterization almost no multilinguality is left (17). Pairing a language with its inversion (i.e., inverted word order) destroys multilinguality as well (3). Having parallel training corpora is helpful for multilinguality (not shown). Results are for embeddings from layer 8.

Dufter and Schütze (2020)

# Multilinguality in LMs

Summary: what affects representation sharing in mBERT and alike?

- Architectural properties (see above)
- Language properties: typological distance, script, word-order similarity, vocabulary overlap
- Setup properties: model capacity relative to the number of languages, data imbalance across languages

# Multilinguality in LMs

Some of these properties relate to variables in linguistics and bilingualism research:

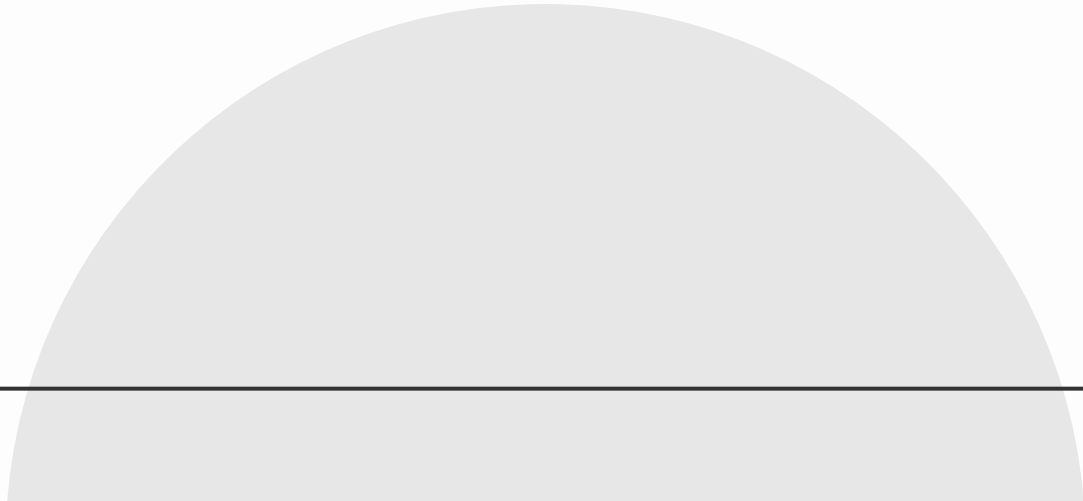
- Language distance - amount of cross-linguistic transfer
- Data amount and order - age of acquisition and exposure
- Capacity per language - proficiency and dominance

But in 'massively multilingual' pretrained models variables are confounded, and we want more control over them.

# Bilingual LMs

- There are (some) pretrained *bilingual* models, and we can evaluate / examine them
  - Zhou & Matushevych (2025) test pretrained monolingual Chinese and bilingual Chinese-English models on Chinese linguistic benchmarks
- But this doesn't solve the issue of controlling for many other variables - ideally we want to train models ourselves.

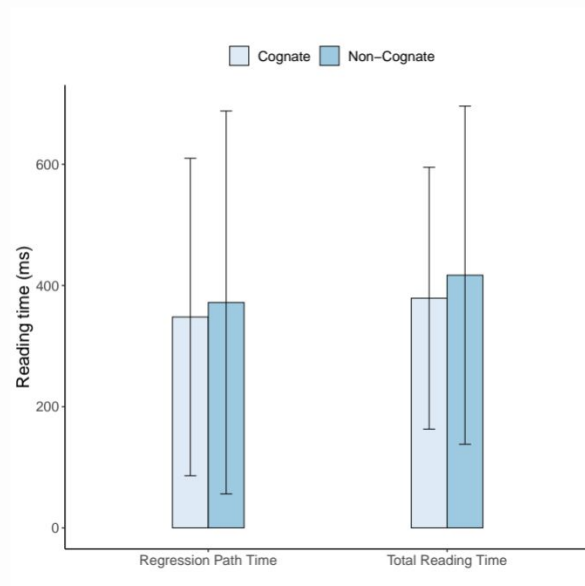
# **Cognate effects**



# Cognate effect

Bultena et al. (2014):

- *The residents disliked the winter / prison ...*
- Dutch translations:
  - winter - winter (cognate)
  - prison - gevangenis (non-cognate control)





# **Cognate effect: causes**

- A. Cumulative frequency hypothesis: Cognate effect just a frequency effect in disguise? (e.g., Strijkers et al., 2010)
- B. Special status of cognates in the human mental lexicon (e.g., van Hell & Dijkstra, 2002)

# Cognate effect in LMs

Cognate processing in an LM allows us to test the cumulative frequency hypothesis

- Cognates are not assigned special status in LMs
- We can count exact word frequencies in training data
- Compare model's predictions to patterns found by Bultena et al. (2014)

# Cognate effect in LMs

Train recurrent models (LSTM) on two languages under various conditions. What all the conditions share:

- First language (L1) Dutch – second language (L2) English
- Training on Wikipedia (2M shuffled sentences per language)
- 80/10/10 training/test/validation

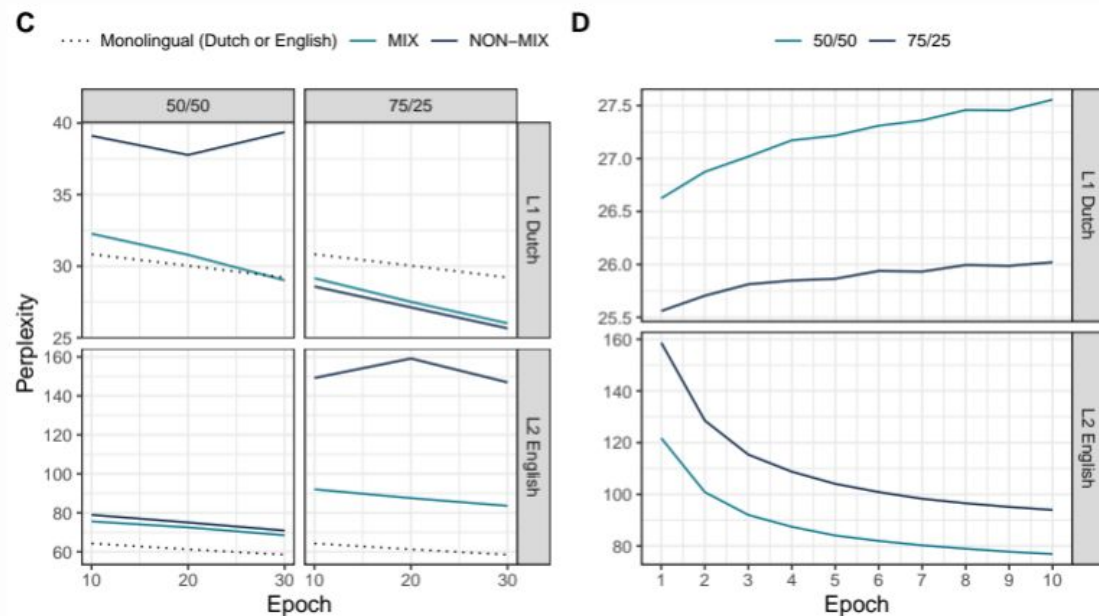
# Cognate effect in LMs

Training setup variables we manipulate:

- **Language mixing**
  - NON-MIX: L1 data followed by L2 data
  - MIX: L1+L2 sentences shuffled
- **L1:L2 training ratio**
  - 75% L1 – 25% L2 data
  - 50% L1 – 50% L2 data
- **L1 pretraining**
  - Yes: 30 epochs on L1 data → 10 epochs on L1+L2 data
  - No: 30 epochs on L1+L2 data

# Sanity check: learning curves

- Perplexity on L1 and L2 test data
- Compare to monolingual English and Dutch LMs
- Cf. catastrophic forgetting

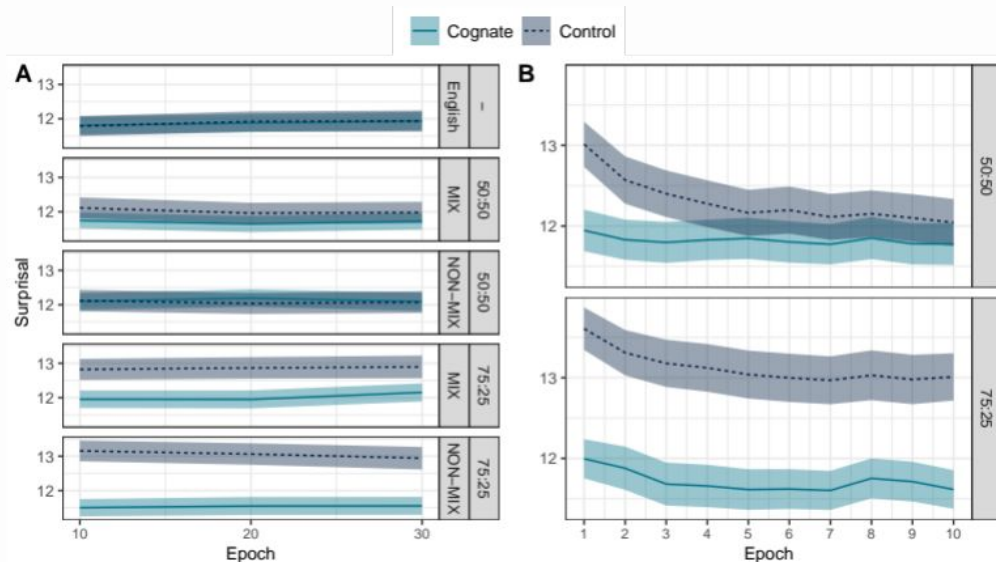


Winther et al. (2021)

# Cognate effect in LMs

Cognate effect observed in some (not all) of the models

We also tested on Norwegian stimuli, not shown



Winther et al. (2021)

# Cognate effect in LMs

- Lower surprisal for cognate compared to control words
- Consistent cognate facilitation effect in two models
- What's common between these models?
  - Higher exposure to L1 (75:25 language ratio)
  - Presentation of L1 before L2 (NON-MIX or PT)

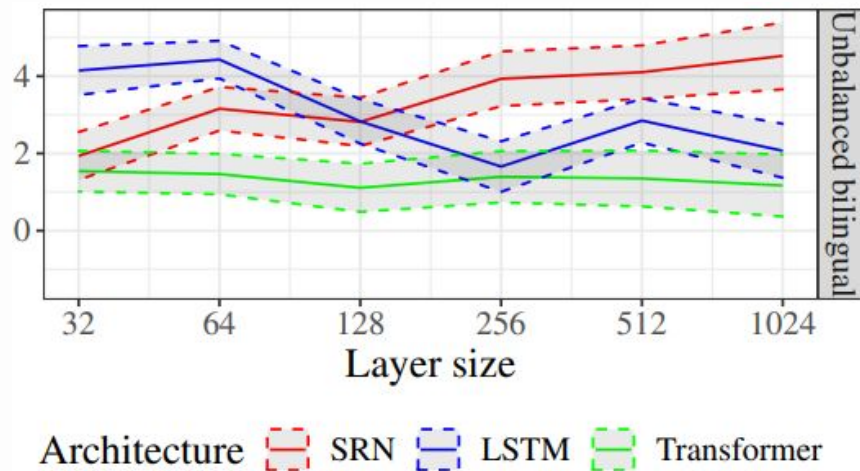
# Cognate effect as frequency effect

- Does word frequency predict the effect better than the (binary) cognate status?
- Fit a series of mixed-effects models predicting word-level surprisal:
  - $\text{surprisal} \sim \text{cognate status}^* \text{epoch} + (1|\text{word}) + (1|\text{sentence}) + (1|\text{initialization})$
  - $\text{surprisal} \sim \text{cumulative frequency}^* \text{epoch} + (1|\text{word}) + (1|\text{sentence}) + (1|\text{initialization})$
- Frequency is a better predictor than cognate status



# Cognate effect across architectures

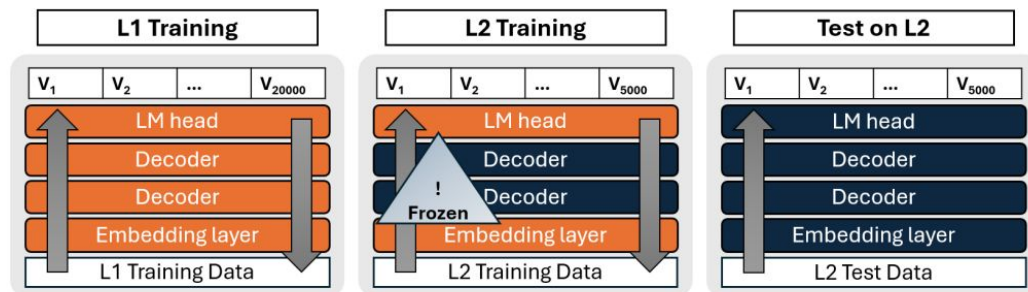
- An analogous study with simpler (SRN) and more complex (transformer) models
- Additionally experimented with hidden layer size (cf. model capacity earlier)



Roslund & Matushevych (2022)

# More advanced manipulations

Aoyama & Schneider  
(2024)



**Figure 1:** Training setup for L2LMs. The model is first pretrained on a given L1. We then freeze all the layers *except for* the embedding and output layers, and then continue pretraining on the L2 (English).

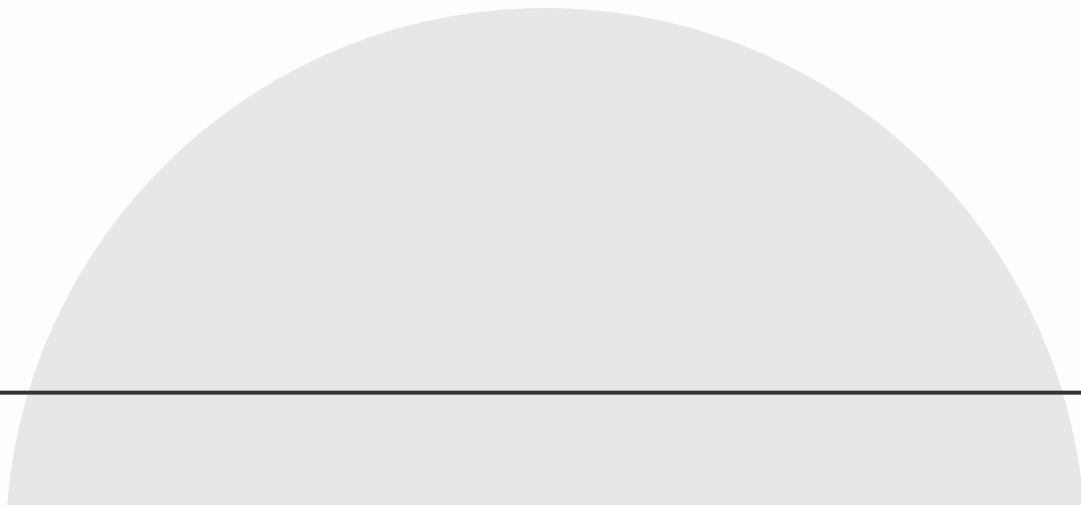
- High-level abstract linguistic knowledge is in the intermediate layers
- Freezing them will force the model to acquire a new language primarily by learning new words while relying on the L1 grammar

# More advanced manipulations

Constantinescu et al (2025)

- Simulating maturational decrease in plasticity (among simulating various other types of bilingual / L2 / heritage speakers)
- Introducing a loss term that penalizes deviations from a prior (L1) distribution

# Priming in LMs



# Structural priming

Syntactic priming: reusing a recently encountered structure

Example: hear/produce a double-object dative, and you're more likely to produce that structure again, even with entirely different words (Bock, 1986)

**PO:** *The nurse handed the chart to the doctor* → PO

**DO:** *The nurse handed the doctor the chart* → DO

## **Structural priming: side note**

- Persistence of structure independent of content is taken as evidence for abstract, lexically-independent syntactic representations (e.g., Pickering & Ferreira, 2008).
- A behavioural window on representation (again)

# Crosslinguistic structural priming

Schoonbaert, Hartsuiker, & Pickering (2007):

Dutch-English bilinguals using dative constructions

- Dutch DO prime: *De jongen geeft het meisje het boek*
- Dutch PO prime: *De jongen geeft het boek aan het meisje*
- English DO target: *The teacher gives the student the pencil*
- English PO target: *The teacher gives the pencil to the student*

# Crosslinguistic structural priming

Bernolet, Hartsuiker, & Pickering (2013):

Dutch-English bilinguals using genitive constructions

- Dutch of-genitive prime: *Het hemd van de jongen is blauw*
- Dutch pronominal s-genitive prime: *De verpleegster haar ei is geel*
- English of-genitive target: *The egg of the nurse is yellow*
- English s-genitive target: *The witch's egg is yellow*



# Crosslinguistic structural priming

Hartsuiker, Pickering, & Veltkamp (2004):

Spanish-English bilinguals using active/passive sentences

- Spanish active prime: *El taxi persigue el camión* (The taxi chases the truck)
- Spanish passive prime: *El camión es perseguido por el taxi* (The truck is chased by the taxi)
- Spanish OVS Prime: *El camión lo persigue un taxi* (The truck, a taxi is chasing it)
- English active target: *The bullet is hitting the bottle*
- English passive target: *The bottle is being hit by the bullet*

# Crosslinguistic structural priming

Fleischer, Pickering, & McLean (2012):

Polish-English bilinguals using active/passive sentences

- Sportowiec przygniata baletnicę. (The sportsman squashes the ballet dancer.)
- Baletnica jest przyginiatana przez sportowca. (The ballet dancer is squashed by the sportsman.)

# Crosslinguistic structural priming

Kotzochampou & Chondrogianni (2022):

Greek-English bilinguals using active/passive sentences

- Ο αθλητής κλωτσάει τον κλέφτη. (The athlete kicks the thief.)
- Ο κλέφτης κλωτσιέται από τον αθλητή. (The thief is kicked by the athlete.)

# Crosslinguistic structural priming

- Robust crosslinguistic priming effects
- Word order should be the same (e.g., wouldn't work with Dutch-English relative clauses because of verb position mismatch)
- Stronger effects in more proficient bilinguals
- Stronger effects in case of lexical overlap
- Effects observed despite morphological differences

# Priming in LMs (within the same language)

Why would there be priming in LMs?

Option A. Sinclair et al. (2022):

- *Structural persistence* in training data: writers and speakers reuse structures → consecutive sentences share structure more often than chance.
- Under this explanation, models trained on shuffled sentences shouldn't exhibit priming

# Priming in LMs (within the same language)

Why would there be priming in LMs?

Option B. Jumelet et al. (2024)

- Priming can be seen as a form of in-context learning
- Not a phenomenon encoded in the data

# Priming in LMs (within the same language)

Zhou et al. (2025): weight updates vs in-context learning

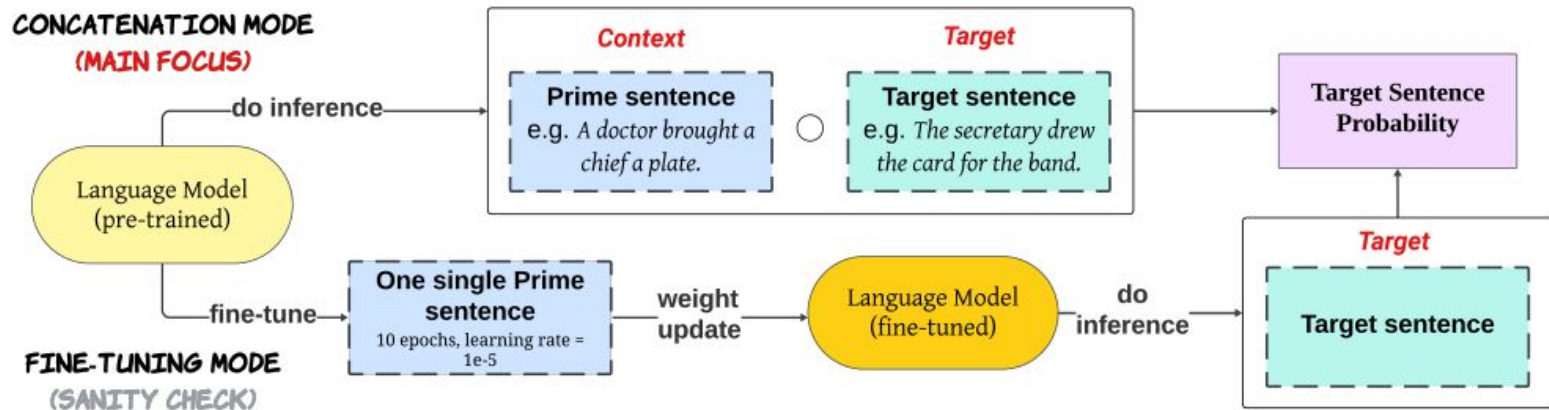


Figure 1: An overview of our experimental design.

# Priming in LMs (within the same language)

Zhou et al. (2025):

- Priming as error-driven (implicit) learning
- Inverse frequency effect: less-preferred syntactic alternatives cause stronger priming than more-preferred structures
  - English *give* is biased towards DO
  - A (less preferred) PO prime with *give* causes a greater priming effect than a DO prime with *give*
- An error signal is implicitly computed in the forward pass of in-context learning



# Priming in LMs (within the same language)

Zhou et al. (2025):

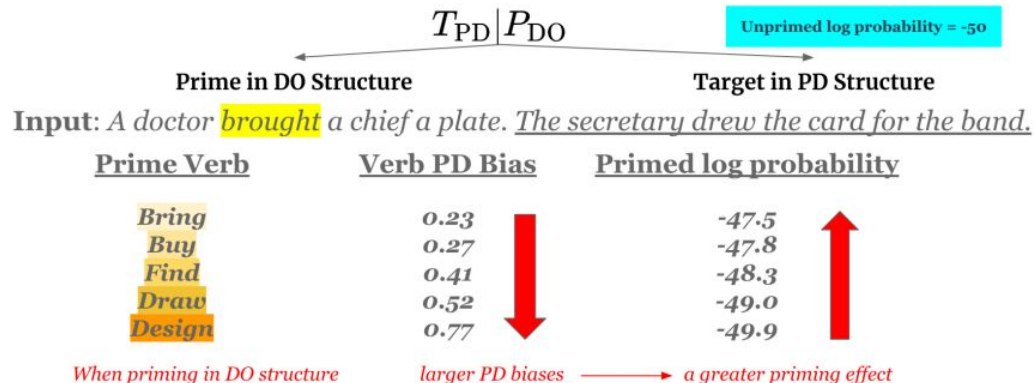


Figure 2: A demonstration of the IFE: a stronger priming effect of a DO prime is predicted as PD-bias increases. The numerical values of the primed log probabilities are hypothetical, for illustration purposes only.

# Priming in LMs (within the same language)

How to compute priming?

Sinclair et al. (2022):

$$PE = \log P(T\_X \mid P\_X) - \log P(T\_X \mid P\_Y)$$

where  $P\_X$  and  $P\_Y$  are the two prime sentences, and  $T\_X$  is the target sentence

# Priming in LMs (within the same language)

How to compute priming?

Arnett et al. (2025):

$$P_{norm}(T\_X | P\_X) = P(T\_X | P\_X) / [P(T\_X | P\_X) + P(T\_Y | P\_X)]$$

vs.

$$P_{norm}(T\_X | P\_Y) = P(T\_X | P\_Y) / [P(T\_X | P\_Y) + P(T\_Y | P\_Y)]$$

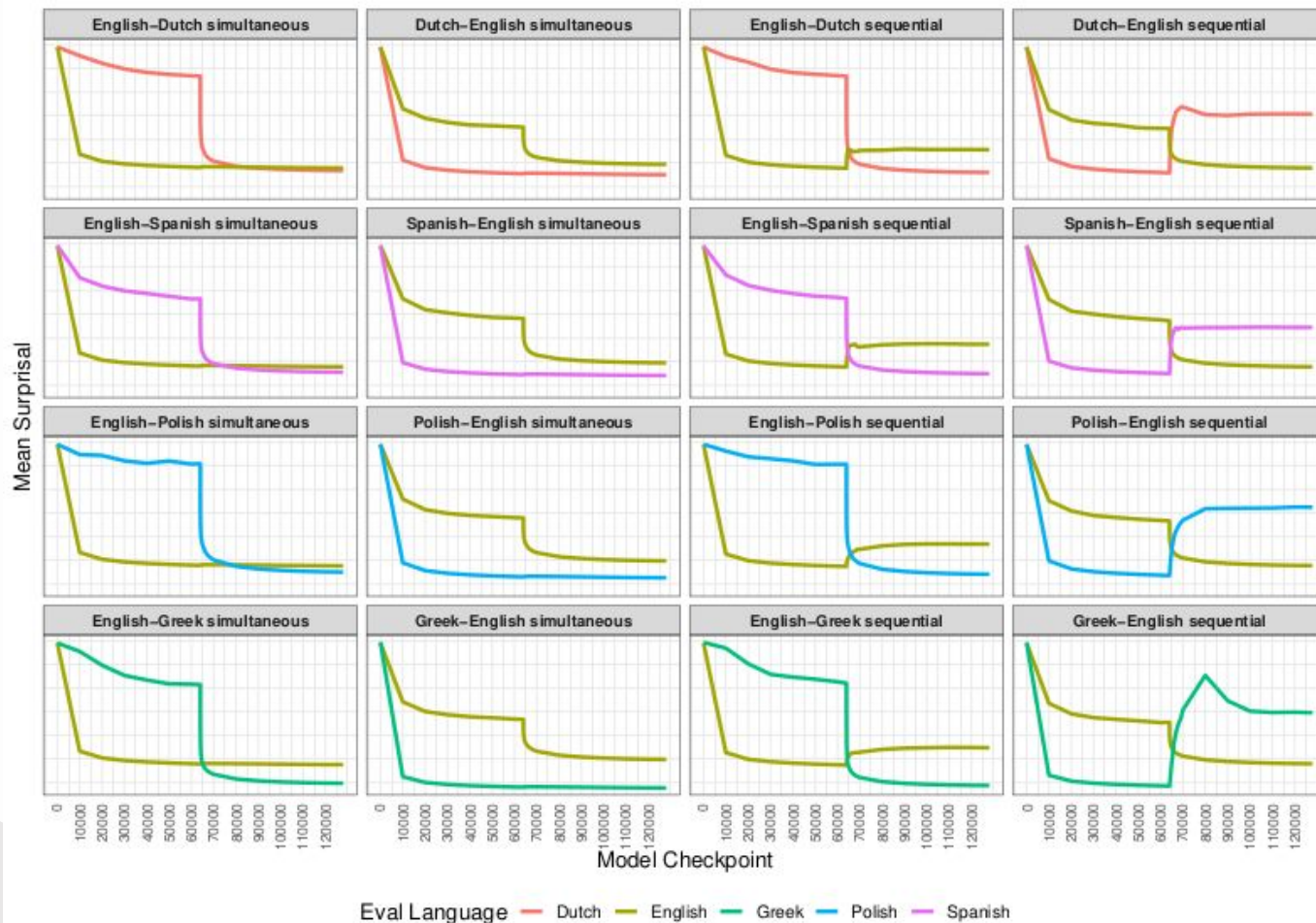
# Cross-lingual priming in LMs

Arnett et al. (2025):

- Train 16 small B-GPT models
- Language pairs: Dutch-English, English-Polish, English-Greek, English-Spanish (in both directions)
- Exposure order: simultaneous vs. sequential
- Per-language data quantity controlled
- Test on 5 data sets mentioned earlier

Arnett et al. (2025)

## Learning curves



## Priming effects

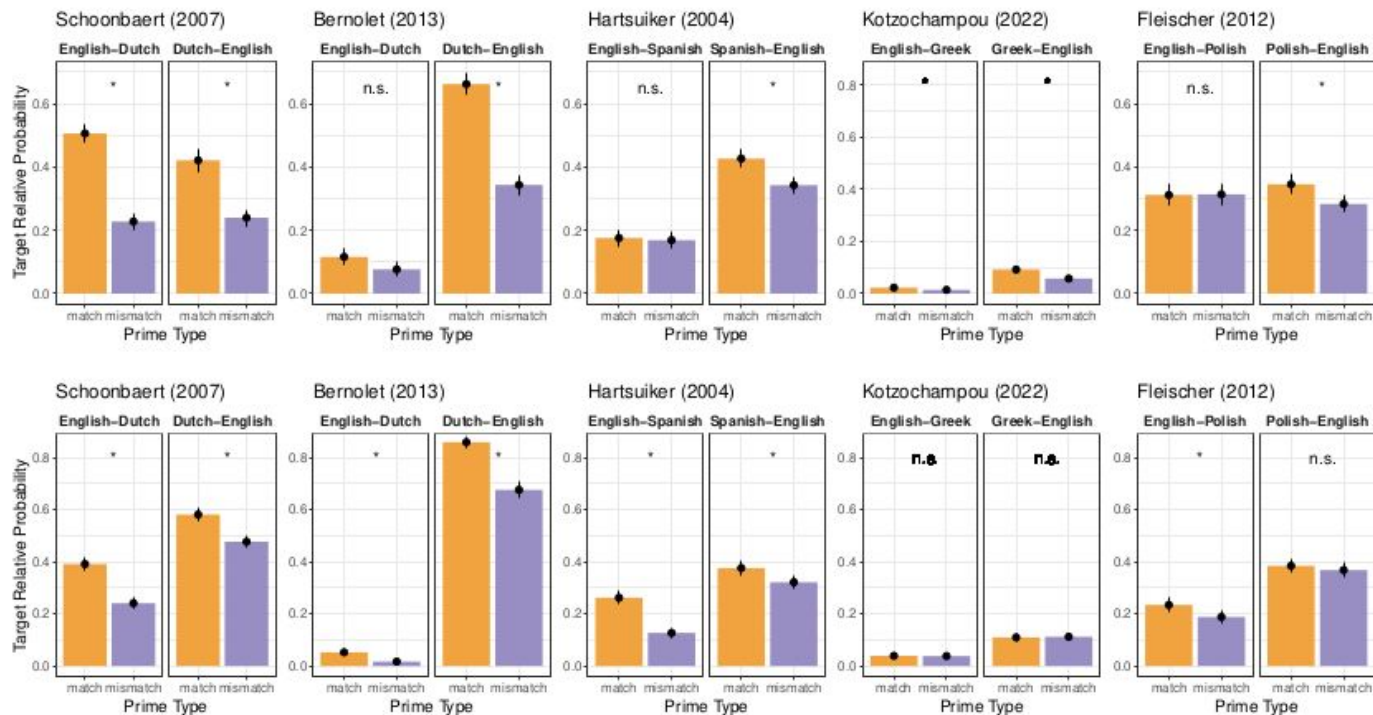


Figure 3: Priming results for the simultaneous (top) and sequential (bottom) bilingual models. For all experiments, prime language corresponds to L1 and target language corresponds to L2. Significance is indicated with \*. Color indicates prime condition. Orange indicates congruent or matching prime and target types and purple indicates mismatched prime and target types. Specific grammatical alternations tested are described in Appendix A.

# Cross-lingual priming in LMs

Arnett et al. (2025):

- Asymmetric across directions - effect more robust when English is the L2 target language (rather than L1 prime language), an informative finding because most human data are on English L2
- Effects are less robust for less similar language pairs

# Unresolved issues in bilingual modeling

- Catastrophic forgetting (and forgetting in general)
  - How to model a good sequential bilingual / heritage speaker?
- Tokenization isn't language-neutral
  - Subword tokenizers capture some word-level similarity patterns, but often not the human-like ones



# LMs as experimental subjects

Arnett et al. (2025):

*“Language models are the first model organism for language researchers (akin to fruit flies in genetics research), in that they offer the possibility to refine hypotheses about language through the manipulation and evaluation of models”*

# Thanks!

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